DOST Meeting Management Technical Specification

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1. Purpose

DOST Meeting Management is a web application for scheduling meetings, collecting invitee availability, confirming final meeting times, managing meeting materials and recordings, maintaining minutes, tracking action items, and showing dashboard notifications.

This specification is written for client review. It describes the delivered system behavior, architecture, data model, API, security requirements, and how the deployment scales. Installation and operating procedures are covered in the separate Deployment Guide.

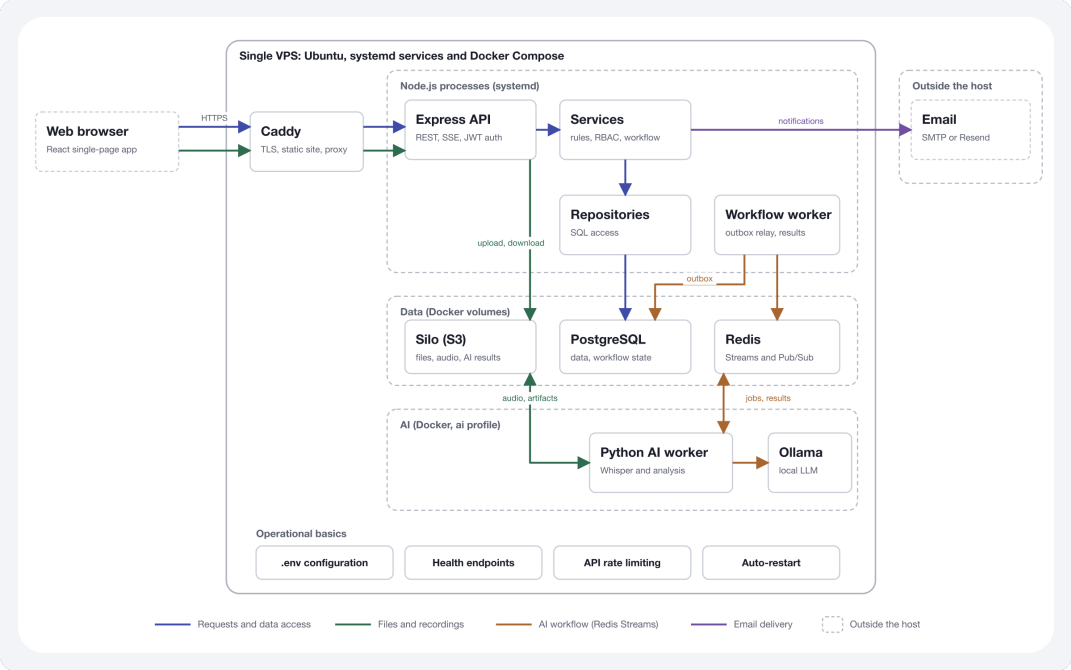
2. User Characteristics

User type	Characteristics	Main goals	System access
Staff user	Authenticated DOST staff member with an email/password account.	View dashboard, access assigned meetings, submit availability, manage assigned action items.	Authenticated web app and API access.
Meeting creator	User who creates a meeting or meeting request.	Create meeting requests, add time options, invite participants, confirm meetings, manage meeting records.	Full control of their created meetings and requests.
Meeting organizer	Participant with an organizer role.	Help manage participants, files, recordings, and minutes for a meeting.	Elevated meeting-scoped permissions.
Attendee	Participant with an attendee role.	View meeting detail, access files/minutes, complete action items.	Meeting-scoped read/collaboration permissions.
IT administrator	Client-side system or infrastructure administrator.	Configure environments, operate deployments, manage capacity, validate service health.	Infrastructure and application administration access.

The system supports general user profile fields such as role and department. Meeting-level authorization is primarily based on meeting creator and participant role.

3. System Architecture

3.1 High-Level Architecture



Caddy is the public browser-facing entrypoint. The browser loads the React application from Caddy and sends same-origin API requests back to Caddy. Caddy then forwards API, SSE, and health requests to the private Express API service. The browser does not call the private API service directly.

The Express API and the Node workflow worker run as systemd services. PostgreSQL, Redis, and Silo object storage run under Docker Compose, and the Python AI worker and Ollama run under the Compose `ai` profile. Email is sent through an external SMTP or Resend account.

3.2 System Components

Component	Technology / approach	Client-facing responsibility
Web application	React single-page application	Dashboard, meeting repository, meeting detail, scheduling, minutes, files, and action items.
API service	Node.js and Express	Business rules, authorization, validation, workflow transitions, file orchestration, SSE.
Node workflow worker	Node.js worker runtime	Dispatches durable workflow outbox events and persists AI result transitions.
Python AI worker	Python, Whisper, Redis Streams	Transcribes recordings and generates structured summaries and proposed action items.
Ollama	Private local LLM runtime	Provides local structured analysis for completed transcripts.

Component	Technology / approach	Client-facing responsibility
Database	PostgreSQL	Authoritative meeting and workflow status, transcripts, analysis results, and outbox events.
Object storage	Silo (S3-compatible, MinIO-compatible fork)	Meeting files, recordings, and generated transcript and analysis artifacts.
Redis	Redis Streams and Pub/Sub	Delivers AI commands/results and broadcasts workflow-status invalidations.
Reverse proxy	Caddy	TLS termination, static web delivery, API routing, SSE, and browser-facing traffic control.
Email	External SMTP or Resend account	Sends meeting invitations, confirmations, and calendar updates.

3.3 Application Modules

Module	Summary
Authentication	Staff login, session validation, password change, authenticated API access.
Dashboard	Upcoming meetings, pending meeting requests, availability requests, action items, notifications.
Meeting repository	Searchable and filterable meeting list with meeting file visibility.
Meeting detail	Meeting metadata, participants, materials, minutes, recording, and action items.
Scheduling requests	Candidate time options, invitee availability, deadline handling, final confirmation.
Participant management	Add/remove participants and manage meeting-scoped participant roles.
File management	Upload, list, download, stream, and delete meeting materials.
Minutes management	Meeting minutes record with editable content and publication fields.
Action items	Meeting-linked and user-linked action-item tracking.
Notifications	In-app feed with read/unread and archive actions; email invitations and calendar updates.
Recording-to-minutes workflow	Durable recording, transcription, analysis, review, and status-update workflow.
AI transcription and summaries	Whisper transcripts, structured AI summaries, and reviewable action-item suggestions.
Administration and RBAC	Centralized account and meeting-role permissions; administrators create and manage accounts.

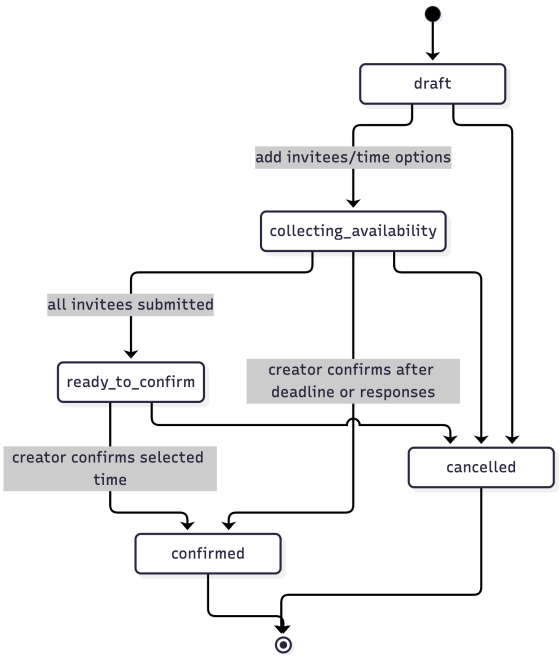
4. Core Workflows

4.1 Authentication

1. User submits email and password.
2. API validates credentials.
3. API returns an authenticated session token and user profile.
4. The web application uses the token for subsequent protected API calls.
5. The API rejects unauthenticated or unauthorized requests.

Self-registration is disabled by default (`SELF_REGISTRATION_ENABLED=false`). Administrators create staff accounts, and the first administrator is created during host setup.

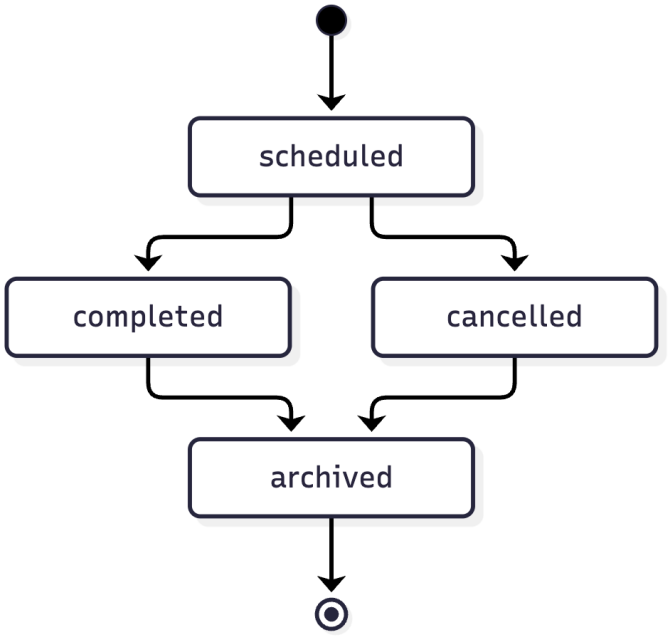
4.2 Meeting Request Scheduling



Important behavior:

- Meeting requests exist before a confirmed meeting is created.
- Invitees submit availability for offered time options.
- The response deadline closes invitee submissions but does not automatically cancel the request.
- The creator may confirm a meeting after the deadline even when some invitees did not respond.
- Confirming creates the scheduled meeting and adds known invitee users as participants.

4.3 Meeting Lifecycle

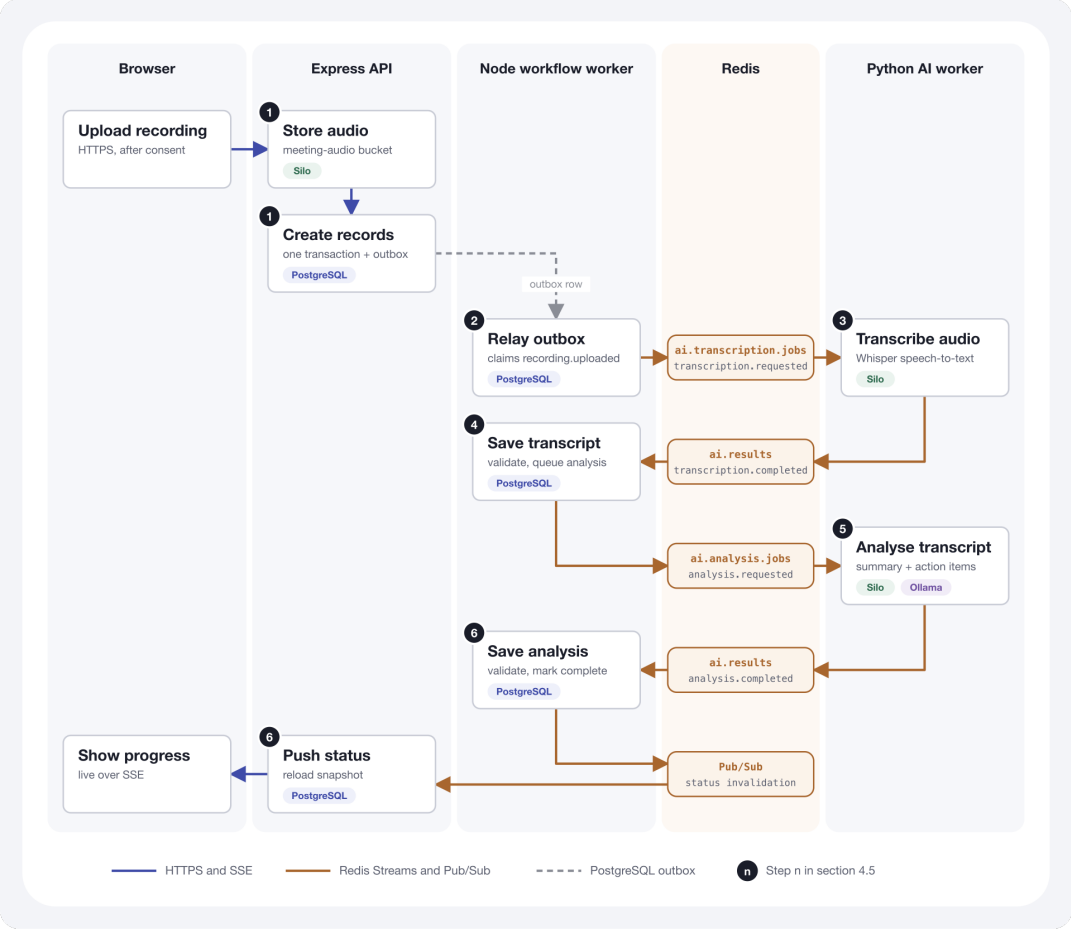


Status transitions are forward-only. Archived meetings are treated as terminal records.

4.4 Files, Recordings, and Minutes

- Meeting materials are uploaded through the web application and stored in object storage.
- Meeting participants can list and download materials for meetings they can access.
- Organizers and creators can manage meeting materials and recordings.
- Download links are short-lived.
- Minutes are stored as a dedicated meeting resource.
- Recording upload creates a durable workflow record when the AI pipeline is enabled.
- Transcripts, AI summaries, decisions, and proposed action items are available for review before they are applied to meeting minutes.

4.5 Recording Analysis Workflow



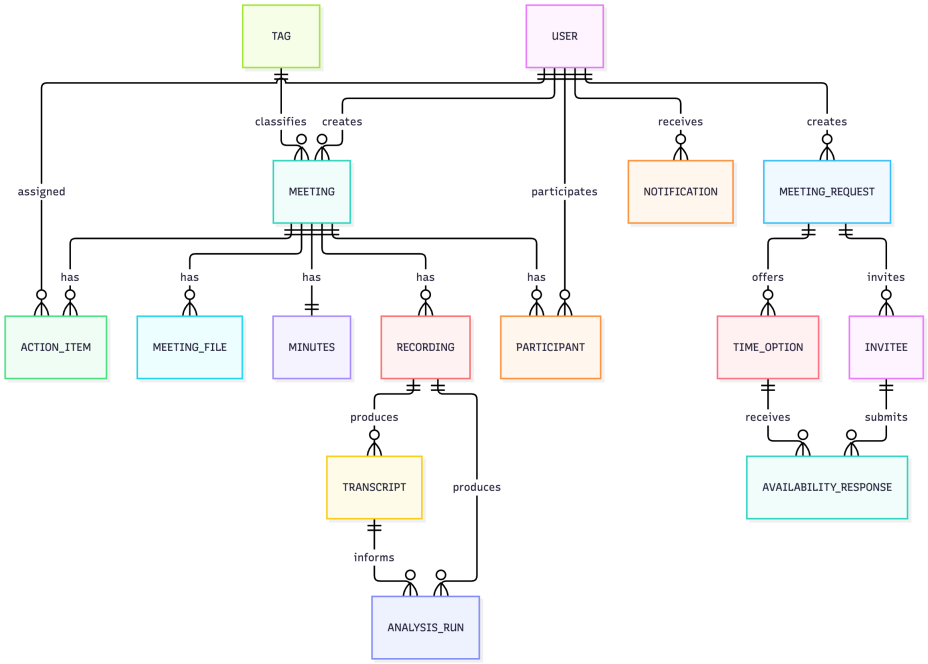
When `AI_PIPELINE_ENABLED=true`, recording processing is asynchronous and uses PostgreSQL as the authoritative workflow state:

1. The API stores the recording in object storage and, in one PostgreSQL transaction, creates a recording, a processing transcript, a queued analysis run, and a `recording.uploaded` workflow-outbox event.
2. The Node workflow worker relays the outbox event to the `ai.transcription.jobs` Redis Stream.
3. The Python AI worker downloads the audio from object storage, runs Whisper, writes a transcript artifact to object storage, and publishes lifecycle/result events to the `ai.results` Redis Stream.
4. The Node workflow worker validates and persists the transcript, then creates and dispatches an `analysis.requested` outbox event.
5. The Python AI worker reads the transcript artifact, calls Ollama for a structured summary, key discussions, decisions, and proposed action items, then publishes the analysis result.
6. The Node workflow worker validates and persists the analysis result, then publishes Redis Pub/Sub invalidations. Connected clients receive refreshed PostgreSQL status snapshots over SSE.

Audio and transcript text are never placed in Redis. Redis carries commands, results, and status invalidations only. Failed or duplicate deliveries are handled through retry settings and guarded PostgreSQL workflow transitions.

5. Database Design

5.1 Entity Relationship Overview



5.2 Logical Data Model

Entity	Purpose	Main relationships
User	Authenticated staff account and profile.	Creates meetings and requests; receives action items.
Meeting	Confirmed or scheduled meeting record.	Has participants, files, minutes, recordings, action items.
Participant	Meeting membership and meeting-scoped role.	Links users to meetings.
Meeting file	Metadata for uploaded meeting materials.	Belongs to one meeting and one uploader.
Action item	Task created from or attached to a meeting.	Belongs to one meeting; may be assigned to one user.
Meeting request	Scheduling poll before a meeting is confirmed.	Has invitees and candidate time options.
Time option	Candidate date/time option for a meeting request.	Receives invitee availability responses.
Invitee	Person invited to respond to a meeting request.	May link to an existing user account.

Entity	Purpose	Main relationships
Availability response	Invitee response for a candidate time option.	Belongs to one invitee and one time option.
Tag	Meeting repository classification.	Classifies meetings.
Recording	Dedicated meeting recording metadata.	Belongs to one meeting; may produce transcripts.
Transcript	Persisted Whisper transcript, segments, provenance, and status.	Belongs to one recording and meeting.
Analysis run	Generated summary, decisions, and reviewable action suggestions.	Uses one transcript and may reference one recording.
Workflow outbox	Durable command awaiting Redis Stream dispatch.	Records recording and analysis workflow events.
Minutes	One meeting minutes record containing prepared meeting notes.	Belongs to one meeting.
Notification	Persisted in-app notification shown in dashboard/topbar contexts.	Belongs to one recipient user and may reference system events.

5.3 Data Integrity Requirements

- Use stable unique identifiers for all primary records.
- Enforce meeting, participant, request, invitee, and recording relationships at the database level.
- Enforce valid status values for meetings, requests, minutes, transcripts, and action items.
- Enforce only one selected scheduling time option per meeting request.
- Enforce only one active dedicated recording per meeting.
- Maintain indexes for common user, meeting, status, date, search, and notification queries.

6. API Interface

All application APIs are exposed under the same browser-facing domain through the reverse proxy. Application paths are prefixed with `/api` unless noted otherwise. Exact request/response schemas should be confirmed during integration and acceptance testing.

The detailed API specification, including endpoint grouping, sample requests, path/query parameters, and expected responses, is maintained in the Postman collection at `docs/postman/dost-meeting-mgnt.postman_collection.json`.

6.1 Authentication and User Session

Method	Path	Resource	Purpose
POST	`/auth/login`	Authentication login	Authenticate a user and issue a session.
GET	`/auth/me`	Current user	Return the current authenticated user.
GET	`/users`	User directory	List users for assignment and participation.

6.2 Meetings

Method	Path	Resource	Purpose
POST	`/meetings`	Meetings	Create a scheduled meeting.
GET	`/meetings`	Meetings	List accessible meetings with pagination, status/date filters, and search.
GET	`/meetings/:id`	Meeting detail	Read one accessible meeting with participants and related records.
PUT	`/meetings/:id`	Meeting detail	Update meeting fields and enforce valid status transitions.
DELETE	`/meetings/:id`	Meeting detail	Delete a meeting where the user has sufficient permission.
GET	`/meetings/:meetingId/participants`	Meeting participants	List participants.
POST	`/meetings/:meetingId/participants`	Meeting participants	Add participants by user and role.
PATCH	`/meetings/:meetingId/participants/:userId`	Meeting participant	Update a participant role.
DELETE	`/meetings/:meetingId/participants/:userId`	Meeting participant	Remove a participant while preserving organizer safeguards.

6.3 Meeting Requests

Method	Path	Resource	Purpose
POST	`/meeting-requests`	Meeting requests	Create a scheduling request.
GET	`/meeting-requests/:requestId`	Meeting request detail	Read request detail, time options, invitees, and status.
GET	`/meeting-requests/:requestId/invitees/:inviteeId`	Invitee request detail	Read invitee-specific request detail.
PUT	`/meeting-requests/:requestId/time-options`	Request time options	Replace candidate time options.

Method	Path	Resource	Purpose
PUT	`/meeting-requests/:requestId/invitees`	Request invitees	Replace invitees.
PUT	`/meeting-requests/:requestId/invitees/:inviteeId/availability`	Invitee availability	Submit availability responses.
POST	`/meeting-requests/:requestId/confirm`	Request confirmation	Confirm selected time and create a meeting.
DELETE	`/meeting-requests/:requestId`	Meeting request	Cancel the request and linked meeting when applicable.

6.4 Files, Recordings, Minutes, and Action Items

Method	Path	Resource	Purpose
POST	`/meetings/:meetingId/files`	Meeting files	Upload a meeting material.
GET	`/meetings/:meetingId/files`	Meeting files	List meeting files.
GET	`/files/:id/download`	File download	Return an authorized download link.
GET	`/files/:id/content`	File content	Stream authorized file content.
DELETE	`/files/:id`	File	Delete file metadata and object storage record.
GET	`/meetings/:meetingId/recording`	Meeting recording	Get the active recording and latest generation status.
PUT	`/meetings/:meetingId/recording`	Meeting recording	Upload or replace a dedicated recording.
DELETE	`/meetings/:meetingId/recording`	Meeting recording	Archive or remove the active recording.
GET	`/meetings/:meetingId/recording/analysis`	Recording analysis	Read the latest transcript analysis for review.
POST	`/meetings/:meetingId/recording/generate`	Generate minutes	Start or retry transcription and analysis.
GET	`/meetings/:meetingId/minutes`	Meeting minutes	Read or initialize meeting minutes.
PUT	`/meetings/:meetingId/minutes`	Meeting minutes	Update meeting minutes.
PUT	`/meetings/:meetingId/minutes/:minutesId`	Meeting minutes	Update a specific minutes record.
POST	`/meetings/:meetingId/minutes/apply-generated`	Meeting minutes	Apply a reviewed AI analysis to the minutes.

Method	Path	Resource	Purpose
GET	`/meetings/:meetingId/action-items`	Meeting action items	List meeting action items.
POST	`/meetings/:meetingId/action-items`	Meeting action items	Create an action item.
GET	`/action-items`	User action items	List current user's action items.
GET	`/action-items/:id`	Action item	Read one action item.
PUT	`/action-items/:id`	Action item	Update action item content, assignee, status, or priority.
DELETE	`/action-items/:id`	Action item	Delete an action item.

6.5 Dashboard, Notifications, Tags, and Health

Dashboard, notification, and tag paths use the `/api` prefix. Health paths are exposed without the `/api` prefix so infrastructure can check service status directly.

Method	Path	Resource	Purpose
GET	`/dashboard`	Dashboard	Return dashboard aggregate data.
GET	`/notifications/summary`	Notification summary	Return latest topbar notification state.
GET	`/notifications`	Notification feed	Return paginated notifications.
POST	`/notifications/:id/read`	Notification action	Mark one notification as read.
POST	`/notifications/read`	Notification action	Mark selected notifications as read.
POST	`/notifications/read-all`	Notification action	Mark all notifications as read.
POST	`/notifications/:id/archive`	Notification action	Archive one notification.
POST	`/notifications/archive-read`	Notification action	Archive all read notifications.
GET	`/tags`	Tags	List meeting tags.
POST	`/tags`	Tags	Create a meeting tag where authorized.
GET	`/health/live`	Liveness health	Confirm the application process is running.
GET	`/health/ready`	Readiness health	Confirm required dependencies are reachable.
GET	`/health`	Readiness health	Return readiness status and process uptime.

7. Security Requirements

7.1 Authentication and Session Security

- Require authenticated sessions for all non-public application functionality.
- Use strong production secrets managed by the deployment environment, not committed files.
- Configure session lifetime according to the client's risk profile.
- Store only hashed passwords.
- Never return password hashes, secret values, or infrastructure credentials in API responses.
- Keep self-registration disabled for staff-only deployments; administrators create accounts.

7.2 Authorization

- Enforce meeting access by creator or participant membership.
- Restrict participant management, file deletion, recording management, and minutes updates to the meeting creator or organizers.
- Restrict meeting request management to the request creator.
- Allow invitee-specific availability submission only for the relevant invitee or request creator.
- Prevent removal or demotion of the last organizer for a meeting.
- Apply account-level (`staff`, `admin`) and meeting-level role permissions from one central permission map; administrators have consistent access across meetings and user management.

7.3 Input Validation

- Validate all external input at the API boundary.
- Validate identifiers, dates, time ranges, status values, roles, priorities, and search parameters.
- Reject malformed or unsupported values early with consistent error responses.
- Sanitize and bound free-text search inputs.
- Apply upload size and file-type validation based on the target upload workflow.

7.4 Network and Proxy Security

- Keep the reverse proxy as the only public entrypoint.
- Keep the API service, database, Redis, and object storage on private network paths.
- Use HTTPS for browser traffic.
- Enable API rate limiting and proxy-aware client IP handling.
- Avoid exposing administrative consoles publicly unless protected by network and identity controls.

7.5 File and Object Storage Security

- Require meeting access before listing, uploading, streaming, or downloading meeting materials.
- Use short-lived download links where presigned access is required.
- Store object storage credentials securely in the deployment environment.
- Separate ordinary meeting files from recording/audio artifacts at the bucket or policy level.
- Treat recording uploads as a high-risk workflow: require uploader consent and enforce the upload size limit.

8. Deployment and Scalability

Installation, configuration, and operating procedures are in the separate Deployment Guide. This section covers only how the current deployment scales.

8.1 Current Deployment

The delivered system runs on a single host (Deployment Guide, Option 1), as shown in Section 3.1. This suits demonstrations and small pilots. It is not highly available: a host or disk failure stops the service. No load test has been run, so no capacity figure is claimed.

8.2 Scalability by Component

Component	Can it scale today?	How it scales
Express API	Yes, with one change	Run more instances behind Caddy. Sessions are stateless JWTs and SSE status updates fan out through Redis Pub/Sub, so any instance can serve any client. Rate limiting is counted in each process's memory and must move to a shared Redis store first.
Node workflow worker	Yes	Run more worker processes. Outbox rows are claimed with `FOR UPDATE SKIP LOCKED`, and AI results are read through a Redis Streams consumer group with a unique consumer name per process.
Python AI worker	Yes	Add worker containers or hosts. Transcription and analysis jobs are shared through Redis Streams consumer groups, and jobs left by a stopped worker are reclaimed. Each worker runs one job at a time.
Ollama	Vertically	Use more CPU cores or a GPU, or move Ollama to its own host through `OLLAMA_BASE_URL`.
PostgreSQL	Vertically	Runs as one Compose container. Scale the host up, or move to a managed or replicated PostgreSQL by changing `DATABASE_URL`.
Redis	Vertically	Runs as one node. Use a managed or Sentinel-based Redis with the `noeviction` policy for high availability.
Object storage	Vertically	Runs as one Silo node. Any S3-compatible store, including managed object storage, can replace it through the `MINIO_*` settings.

Recording and file uploads are buffered in API memory (up to 50 MB per request), so API memory should grow with the number of concurrent uploads.

8.3 Scaling Path

1. Scale the single host up: more vCPU and RAM, or a GPU for Whisper and Ollama.
2. Move PostgreSQL and object storage to managed services and keep one application host (Deployment Guide, Option 2).
3. Run several API, workflow-worker, and AI-worker instances on separate hosts or on Kubernetes or OpenShift (Deployment Guide, Option 3). This needs the shared rate-limit store and container images for the Node.js services, which do not exist yet.

9. Testing and Validation

Scope	Validation expectation
Functional testing	Login, dashboard, meeting creation, scheduling requests, meeting detail, files, minutes.
API testing	Authentication, authorization, validation errors, pagination, status transitions.
Database testing	Migration application, data integrity constraints, relationship behavior.
File testing	Upload, download, stream, delete, size limits, unauthorized access rejection.
Security testing	Authenticated route protection, meeting-scoped permissions, rate limiting.
Deployment smoke testing	Liveness/readiness health, login, dashboard, core meeting workflow, file access.
Load testing	Dashboard, meeting repository search, meeting detail, uploads, and action-item updates.

Readiness health should fail when required dependencies are unavailable. Liveness health should only confirm that the application process is running.